Week #9 : Ecological Systems - Reproductive Strategies, Synthesis, Assessment & Biodiversity

Subject/الموضوع: Pre-AP Biology Unit Title: Ecological Systems Grade9: الصف

| Lessons الحصص | Learning targets أهداف التعلم | Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز | Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور |
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| Lesson 1: الحصة 1 | I can compare r-selected and K-selected reproductive strategies.<br>I can use lifespan, maturity, offspring number and parental care to justify a strategy prediction. | 1.1 Flexible display of information (size, color, font, layout). | Lesson 13: Reproductive Strategies — r/K continuum, species evidence and UAE desert examples.<br>Review Teams / Biology / General channel |
| Lesson 2 الحصة الثانية | I can connect matter cycling, energy flow and population dynamics within one ecosystem.<br>I can construct annotated pathways and explain links between ECO 1 and ECO 2. | 5.3 Build fluency with graduated scaffolds and models. | Lesson 14: ECO 1 + ECO 2 Synthesis — systems sort, concept map and matter/energy pathway models.<br>Review Teams / Biology / General channel |
| Lesson 3 الحصة الثالثة | I can interpret ecological data and build an evidence-based explanation of ecosystem change.<br>I can connect rainfall, producer biomass, energy availability and carrying capacity in a UAE reserve case. | 6.3 Facilitate organization with graphic organizers, prompts, and note-taking guides. | Lesson 14: ECO 1 + ECO 2 Learning Checkpoint — UAE reserve CER, AP-style checkpoint parts A and B.<br>Review Teams / Biology / General channel |
| Lesson 4 الحصة الرابعة | I can demonstrate my understanding of ECO 1 and ECO 2 concepts in an independent summative assessment.<br>I can interpret models, data and ecological relationships accurately. | 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. | Summative Test — ECO 1 Cycling of Matter and ECO 2 Population Dynamics; assessment paper and approved materials. |
| Lesson 5 الحصة الخامسة | I can explain how biodiversity supports ecosystem processes and resilience.<br>I can describe ecosystem services and use evidence to predict effects of biodiversity loss. | 8.2 Vary challenge level with differentiated tools, resources, and difficulty. | Lesson 15: Biodiversity, Ecosystem Stability & Services — resilience data and UAE mangrove case study.<br>Review Teams / Biology / General channel |

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| **Teacher's notes/ Reminders**<br>**ملاحظات المعلم والتذكير** | • Bring Biology notebook and calculator for review and data interpretation.<br>• Thursday: Summative Test; follow school assessment procedures and permitted-materials guidance.<br>• Use Teams for the PPTs and checkpoint review; resume with biodiversity after the test. |

الأسبوع# Week #9

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| **Technology Integration** دمج التكنولوجيا | • PowerPoint diagrams, ECO 1/ECO 2 models, data tables and graphs.<br>• Microsoft Teams for lesson resources and revision.<br>• Period 4 summative assessment is conducted independently. |
| **Cross-curricular links** الروابط بين المواد الدراسية | • Mathematics: percentages, population data and graph interpretation.<br>• Geography/Environmental Science: UAE rainfall, drought, reserves and mangroves.<br>• Scientific literacy: claim-evidence-reasoning and systems models. |
| **External Assessment Integration** دمج التقييمات الخارجية | Pre-AP Biology Unit 1: Ecological Systems — ECO 1 Cycling of Matter; ECO 2 Population Dynamics; ECO 3 Importance of Biodiversity. Period 4: summative assessment of previously studied ECO 1 and ECO 2 content. |
| **Wellbeing/ ATL Skills** الصحة النفسية/ أساليب التعلم | • Thinking: systems thinking, comparison, data interpretation and evidence-based inference.<br>• Communication: CER writing, models and explanations.<br>• Self-management: independent summative test and revision. |
| **Standards** المعايير | • Lesson 1: ECO 2.2.2(a–b), reproductive strategies.<br>• Lessons 2–3: ECO 1 + ECO 2 Learning Checkpoint 1 — Cycling of Matter and Population Dynamics.<br>• Lesson 4: Summative Test — ECO 1 + ECO 2 (previously taught content).<br>• Lesson 5: ECO 3.1(a), ECO 3.1(b); EK ECO 3.1.1a–c — Importance of Biodiversity. |
| **Inquiry/ TOK Question** البحث والاستقصاء أسئلة TOK | Lesson 1: Why can different reproductive strategies both be successful?<br>Lesson 2: How are matter cycling, energy flow and populations connected?<br>Lesson 3: How can drought affect several parts of an ecosystem at once?<br>Lesson 4: How well can I independently apply ECO 1 and ECO 2 concepts?<br>Lesson 5: Why can losing species change how an ecosystem functions? |
| **Success Criteria** معايير النجاح **(Measurable)** | Lesson 1 — Reproductive Strategies<br>MUST: Identify r- and K-associated traits.<br>SHOULD: Compare offspring number, lifespan, maturity and parental care.<br>COULD: Justify a strategy prediction using life-history evidence.<br><br>Lesson 2 — ECO 1 + ECO 2 Synthesis<br>MUST: Identify key concepts from ECO 1 and ECO 2.<br>SHOULD: Use annotated pathways to explain matter and energy.<br>COULD: Connect cycles, energy and population change in one model.<br><br>Lesson 3 — Learning Checkpoint<br>MUST: Identify relevant concepts and data patterns.<br>SHOULD: Explain the UAE reserve data using evidence.<br>COULD: Construct a CER connecting drought, plants, energy and populations. |

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| | Lesson 4 — Summative Test<br>MUST: Recall accurate ECO 1 and ECO 2 concepts.<br>SHOULD: Interpret data and ecological models independently.<br>COULD: Justify multistep ecological explanations with evidence.<br><br>Lesson 5 — Biodiversity & Ecosystem Services<br>MUST: Define biodiversity and identify ecosystem services.<br>SHOULD: Explain biodiversity's role in food webs, cycling and resilience.<br>COULD: Justify how biodiversity loss affects stability and human benefits. |
| **National Identity Domain**<br>**مجالات الربط بالهوية الوطنية** | Conservation الحفاظ علي البيئة Respect الاحترام Heritage التراث |
| | المجال Area: Culture / Sustainability / Conservation |
| | • UAE desert species illustrate different reproductive strategies under variable rainfall.<br>• UAE reserve drought data connect water availability, plants, energy and oryx populations.<br>• UAE mangrove communities illustrate biodiversity, carbon storage, water filtration and coastal protection. |